

OrderPulse — Step 2: Protobufs & gRPC Definitions

Technical Instruction & Execution Blueprint for AI Developer Agents

OBJECTIVE

Define gRPC service contracts using Protocol Buffers (.proto) for internal communication between Order Service and Inventory Service, and configure the Go code compilation pipeline.

Step 2 Tasks

- **1. Directory Layout:** Create versioned proto folders: `proto/inventory/v1/` and `proto/order/v1/`.
- **2. Inventory Spec:** Create `inventory.proto` defining `CheckStock` and `ReserveStock` RPC methods.
- **3. Order Spec:** Create `order.proto` defining `CreateOrder` and `GetOrderStatus` RPC methods.
- **4. Generation Pipeline:** Setup generation script/Makefile target using `protoc` with Go plugins.
- **5. Verification:** Confirm generated `*.pb.go` and `*_grpc.pb.go` code files compile cleanly with `go mod tidy`.

Protocol Buffer Specifications

proto/inventory/v1/inventory.proto

```
syntax = "proto3";

package inventory.v1;

option go_package = "github.com/orderpulse/proto/inventory/v1;inventoryv1";

service InventoryService {
  rpc CheckStock (CheckStockRequest) returns (CheckStockResponse);
  rpc ReserveStock (ReserveStockRequest) returns (ReserveStockResponse);
}

message CheckStockRequest {
  string product_id = 1;
  int32 quantity = 2;
}

message CheckStockResponse {
  bool is_available = 1;
  int32 available_quantity = 2;
  double price = 3;
}

message ReserveStockRequest {
  string order_id = 1;
  string product_id = 2;
  int32 quantity = 3;
}

message ReserveStockResponse {
  bool success = 1;
  string message = 2;
}
```

proto/order/v1/order.proto

```
syntax = "proto3";

package order.v1;

option go_package = "github.com/orderpulse/proto/order/v1;orderv1";

service OrderService {
  rpc CreateOrder (CreateOrderRequest) returns (CreateOrderResponse);
  rpc GetOrderStatus (GetOrderStatusRequest) returns (GetOrderStatusResponse);
}

message CreateOrderRequest {
  string user_id = 1;
  string product_id = 2;
  int32 quantity = 3;
}

message CreateOrderResponse {
  string order_id = 1;
  string status = 2;
  double total_amount = 3;
}

message GetOrderStatusRequest {
  string order_id = 1;
}

message GetOrderStatusResponse {
  string order_id = 1;
  string status = 2;
  string created_at = 3;
}
```

Verification Commands

```
# Run generation script
./scripts/gen-proto.sh

# Verify output files
ls -la proto/inventory/v1/*.pb.go
ls -la proto/order/v1/*.pb.go

# Resolve Go dependencies
go mod tidy
```